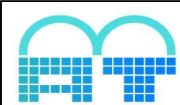
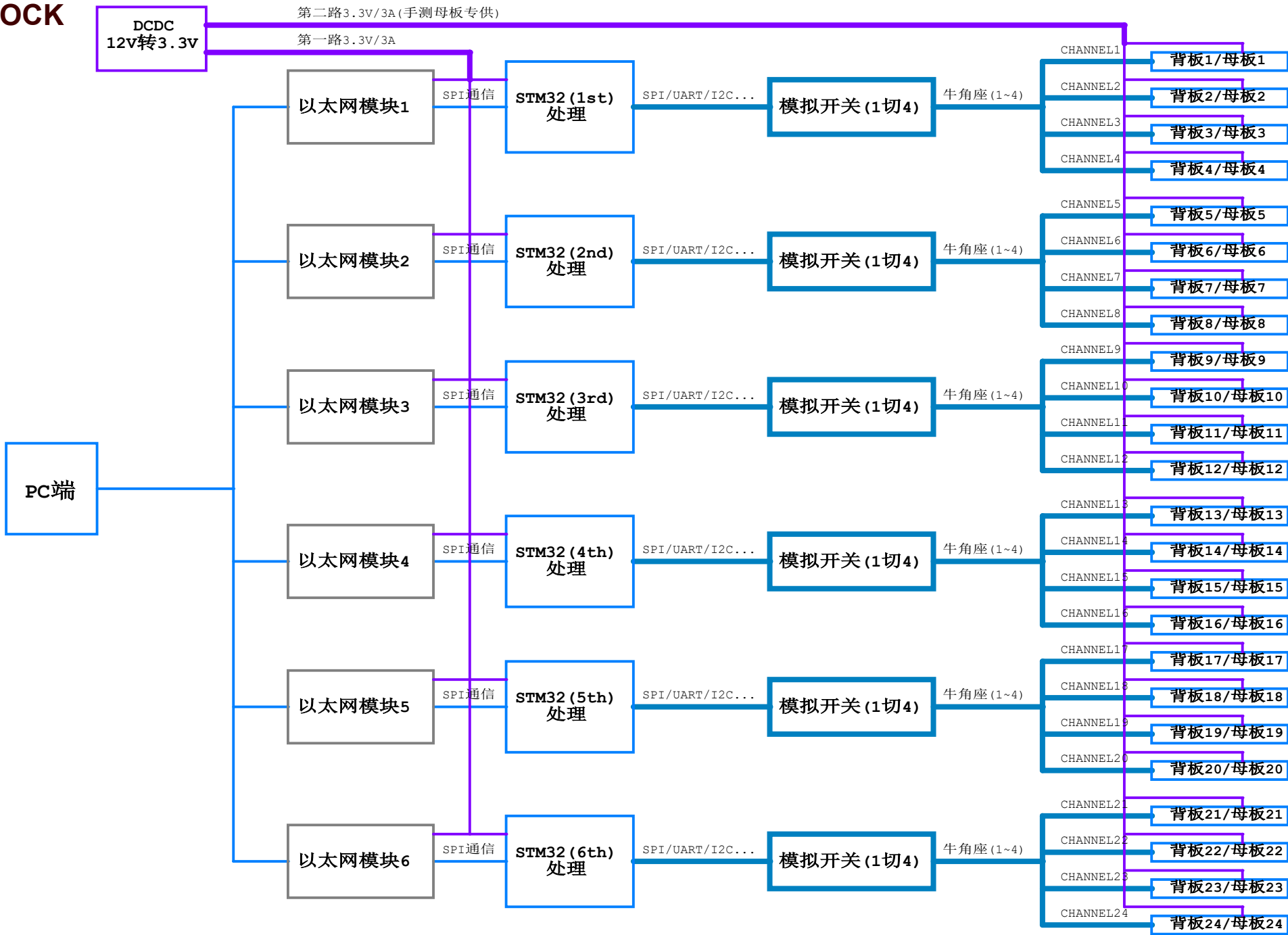


# VERSION HISTORY

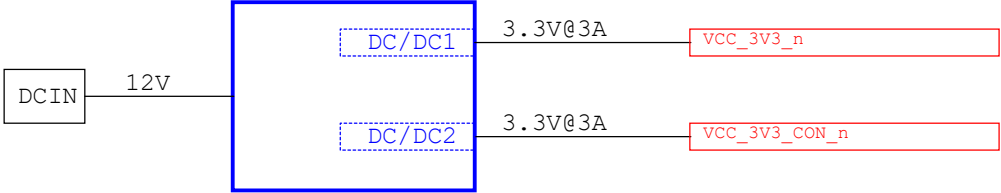
## Index:

- P01 VERSION HISTORY
- P02 BLOCK DIAGRAM
- P03 POWER TREE
- P04 GPIO ASSIGNMENT
- P05 LAYOUT REQUIREMENT
- P06 STM32MINSYS\_W5500\_1
- P07 STM32MINSYS\_W5500\_2
- P08 STM32MINSYS\_W5500\_3
- P09 STM32MINSYS\_W5500\_4
- P10 STM32MINSYS\_W5500\_5
- P11 STM32MINSYS\_W5500\_6

| Revision | Description                                                    | Date     | Drawn | Checked | Approved |
|----------|----------------------------------------------------------------|----------|-------|---------|----------|
| Ver 1.0  | Releas version                                                 | 20220906 | LXJ   |         |          |
| Ver 2.0  | 1: 修改了STM32的BOOT电路<br>2: 删除了P07到P11的JTAG电路<br>3: 增加了SWDIO的上拉电阻 | 20220909 | LXJ   |         |          |



POWER TREE



NC

NC

# GPIO ASSIGNMENT

## THE FIRST STM32(P06)

| PIN  | Define    | CFG | Function |
|------|-----------|-----|----------|
| PA0  |           |     |          |
| PA1  |           |     |          |
| PA2  |           |     |          |
| PA3  |           |     |          |
| PA4  | /SPI1_SCS |     |          |
| PA5  | SPI1_SCLK |     |          |
| PA6  | SPI1_MISO |     |          |
| PA7  | SPI1_MOSI |     |          |
| PA8  | PA8-S     |     |          |
| PA9  | MCU-UTX   |     |          |
| PA10 | MCU-URX   |     |          |
| PA11 | PA11      |     |          |
| PA12 | PA12      |     |          |
| PA13 | SWDIO     |     |          |
| PA14 | SWCLK     |     |          |
| PA15 | TDI       |     |          |

| PIN  | Define    | CFG | Function |
|------|-----------|-----|----------|
| PB0  |           |     |          |
| PB1  |           |     |          |
| PB2  | BOOT1     |     |          |
| PB3  | TD0       |     |          |
| PB4  | TRST      |     |          |
| PB5  |           |     |          |
| PB6  |           |     |          |
| PB7  |           |     |          |
| PB8  |           |     |          |
| PB9  |           |     |          |
| PB10 | MCU-SCL   |     |          |
| PB11 | MCU-SDA   |     |          |
| PB12 | SPI2-CS   |     |          |
| PB13 | SPI2-CLK  |     |          |
| PB14 | SPI2-MISO |     |          |
| PB15 | SPI2-MOSI |     |          |

| PIN  | Define   | CFG | Function |
|------|----------|-----|----------|
| PC0  |          |     |          |
| PC1  |          |     |          |
| PC2  |          |     |          |
| PC3  |          |     |          |
| PC4  | /INT     |     |          |
| PC5  | /RESET   |     |          |
| PC6  | PC6-S    |     |          |
| PC7  | PC7-S    |     |          |
| PC8  |          |     |          |
| PC9  |          |     |          |
| PC10 | UART4-TX |     |          |
| PC11 | UART4-RX |     |          |
| PC12 |          |     |          |
| PC13 |          |     |          |
| PC14 |          |     |          |
| PC15 |          |     |          |

## THE SECOND THROUGH THE SIXTH STM32s(P07-P11)

| PIN  | Define    | CFG | Function |
|------|-----------|-----|----------|
| PA0  |           |     |          |
| PA1  |           |     |          |
| PA2  |           |     |          |
| PA3  |           |     |          |
| PA4  | /SPI1_SCS |     |          |
| PA5  | SPI1_SCLK |     |          |
| PA6  | SPI1_MISO |     |          |
| PA7  | SPI1_MOSI |     |          |
| PA8  | PA8-S     |     |          |
| PA9  | MCU-UTX   |     |          |
| PA10 | MCU-URX   |     |          |
| PA11 | PA11      |     |          |
| PA12 | PA12      |     |          |
| PA13 | SWDIO     |     |          |
| PA14 | SWCLK     |     |          |
| PA15 |           |     |          |

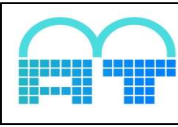
| PIN  | Define    | CFG | Function |
|------|-----------|-----|----------|
| PB0  |           |     |          |
| PB1  |           |     |          |
| PB2  | BOOT1     |     |          |
| PB3  |           |     |          |
| PB4  |           |     |          |
| PB5  |           |     |          |
| PB6  |           |     |          |
| PB7  |           |     |          |
| PB8  |           |     |          |
| PB9  |           |     |          |
| PB10 | MCU-SCL   |     |          |
| PB11 | MCU-SDA   |     |          |
| PB12 | SPI2-CS   |     |          |
| PB13 | SPI2-CLK  |     |          |
| PB14 | SPI2-MISO |     |          |
| PB15 | SPI2-MOSI |     |          |

| PIN  | Define   | CFG | Function |
|------|----------|-----|----------|
| PC0  |          |     |          |
| PC1  |          |     |          |
| PC2  |          |     |          |
| PC3  |          |     |          |
| PC4  | /INT     |     |          |
| PC5  | /RESET   |     |          |
| PC6  | PC6-S    |     |          |
| PC7  | PC7-S    |     |          |
| PC8  |          |     |          |
| PC9  |          |     |          |
| PC10 | UART4-TX |     |          |
| PC11 | UART4-RX |     |          |
| PC12 |          |     |          |
| PC13 |          |     |          |
| PC14 |          |     |          |
| PC15 |          |     |          |

## ANALOG SWITCH LOGIC

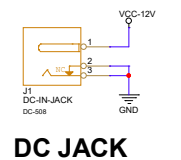
| INPUT |       |       | SWITCH        |
|-------|-------|-------|---------------|
| PA8-S | PC7-S | PC6-S |               |
| 1     | X     | X     | /             |
| 0     | 0     | 0     | CH0 (DEFAULT) |
| 0     | 0     | 1     | CH1           |
| 0     | 1     | 0     | CH2           |
| 0     | 1     | 1     | CH3           |

LAYOUT REQUIREMENT(NOW NOT)



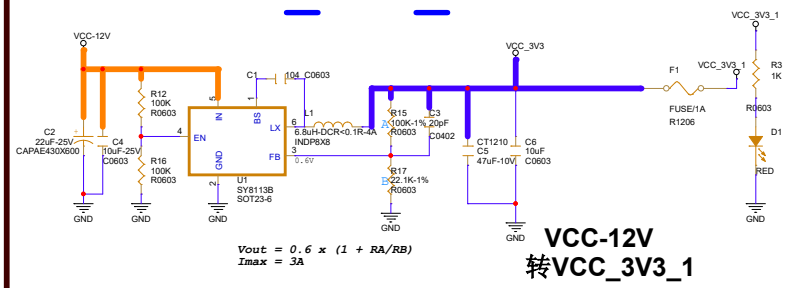
|                                   |                            |               |
|-----------------------------------|----------------------------|---------------|
| Artmem Technology Co.,Ltd         |                            |               |
| Design Name                       |                            |               |
| A100_PER3_LPDDR4_32X1_AXP707_V1_0 |                            |               |
| Size                              | Page Name                  | Rev           |
| A3                                | LAYOUT REQUIREMENT         | 02            |
| Date:                             | Friday, September 09, 2022 | Sheet 5 of 11 |

电源输入



DC JACK

VCC\_3V3\_1

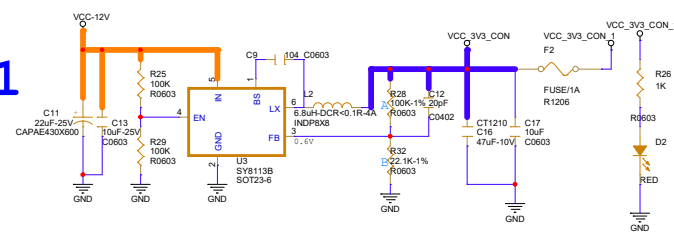


$V_{out} = 0.6 \times (1 + RA/RB)$   
 $I_{max} = 3A$

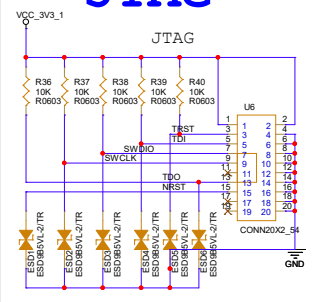
VCC-12V  
转VCC\_3V3\_1

VCC\_3V3\_CON\_1

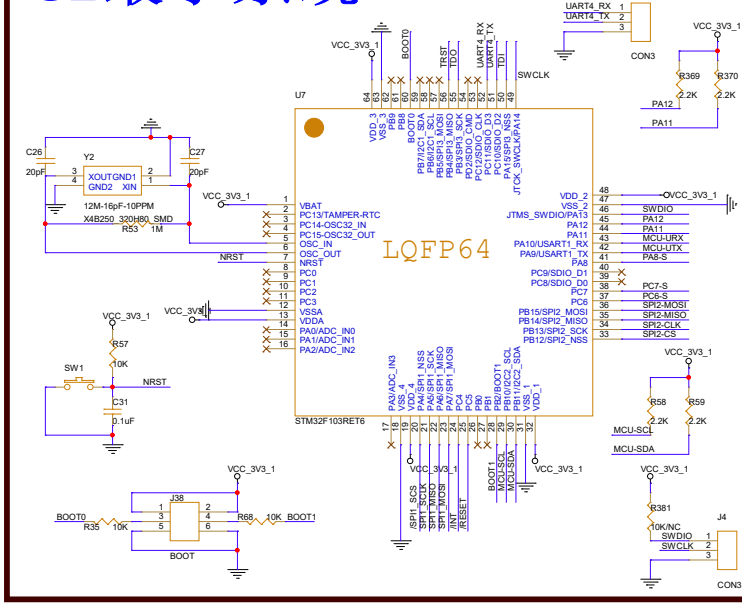
$V_{out} = 0.6 \times (1 + RA/RB)$   
 $I_{max} = 3A$



JTAG

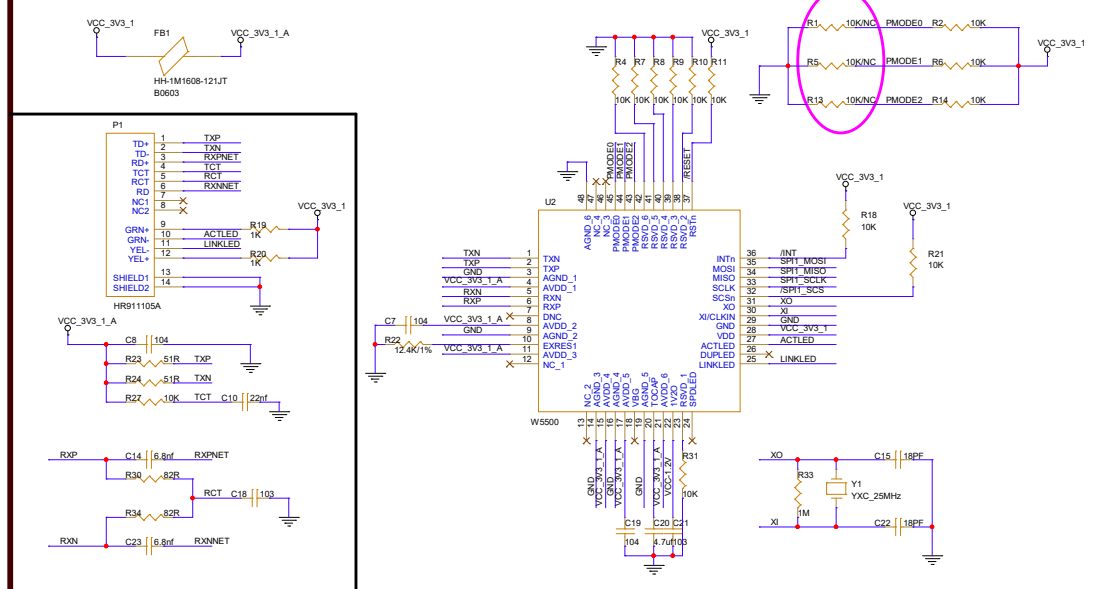


32最小系统

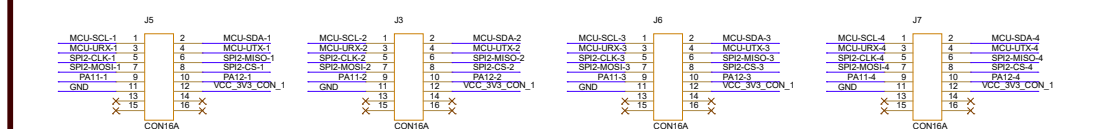


LQFP64

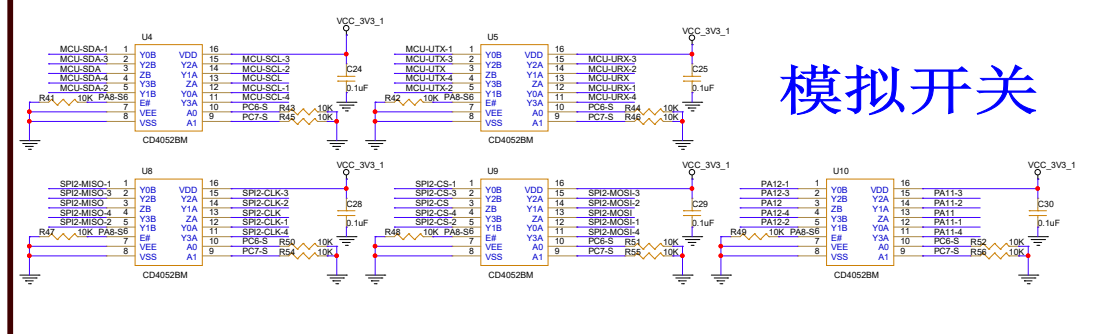
W5500



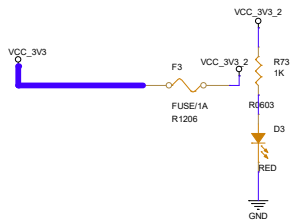
DC牛角座



模拟开关

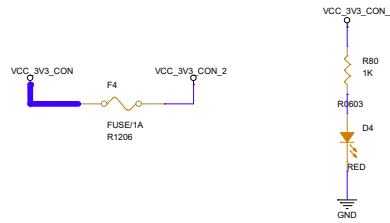


## VCC\_3V3\_2

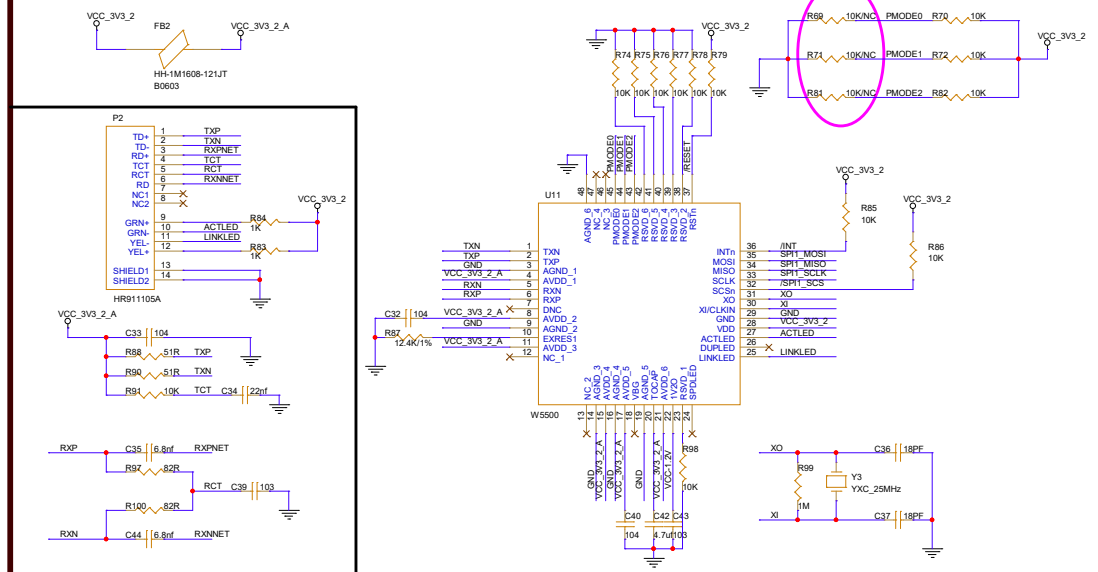


VCC-12V  
转VCC\_3V3\_2

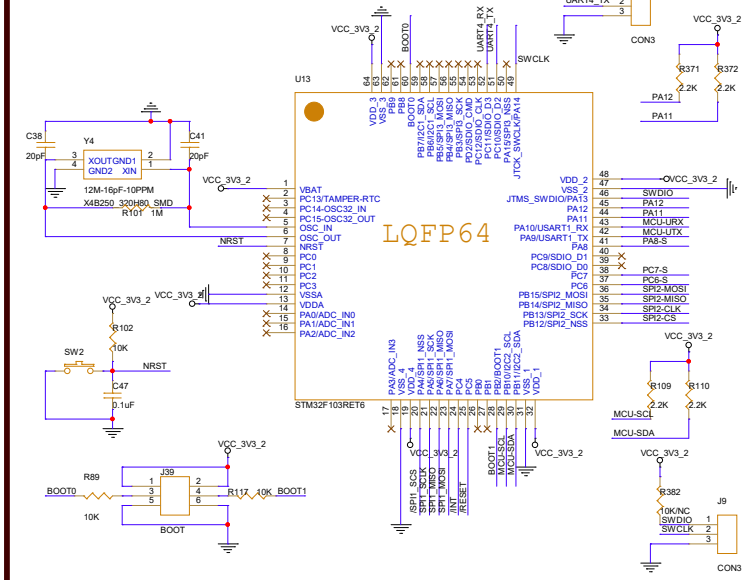
## VCC\_3V3\_CON\_2



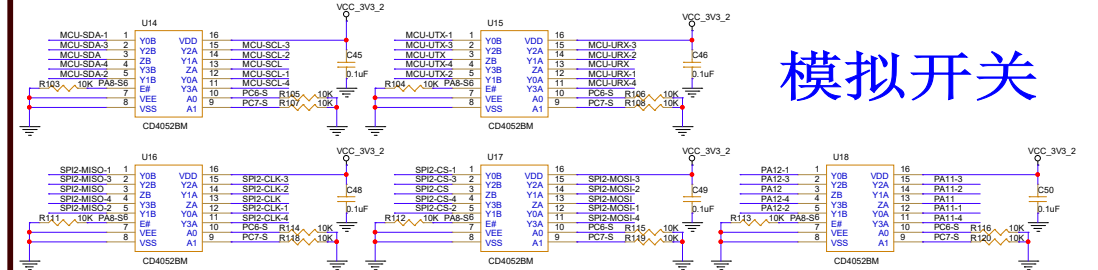
## W5500



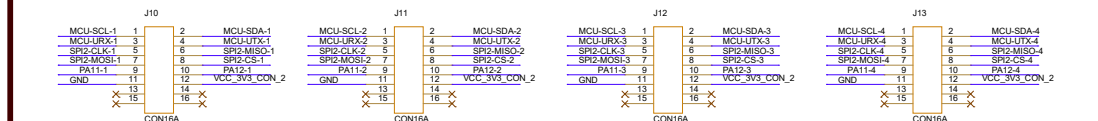
## 32最小系统



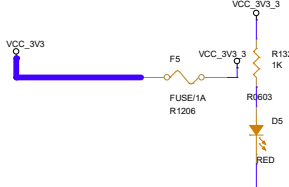
## 模拟开关



## DC牛角座



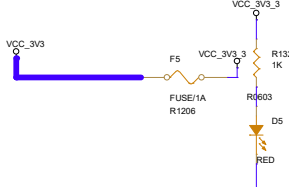
# VCC\_3V3\_3



The diagram shows a circuit for VCC\_3V3\_3. It starts with a VCC\_3V3 input on the left, which connects to a 1A fuse (F5) and a 1K resistor (R132). The output of the fuse and resistor is labeled VCC\_3V3\_3. This output is connected to a 603 resistor (R603) and a red LED (D5). The LED is connected to ground (GND).

**VCC-12V**  
**转VCC\_3V3\_3**

# VCC\_3V3\_3

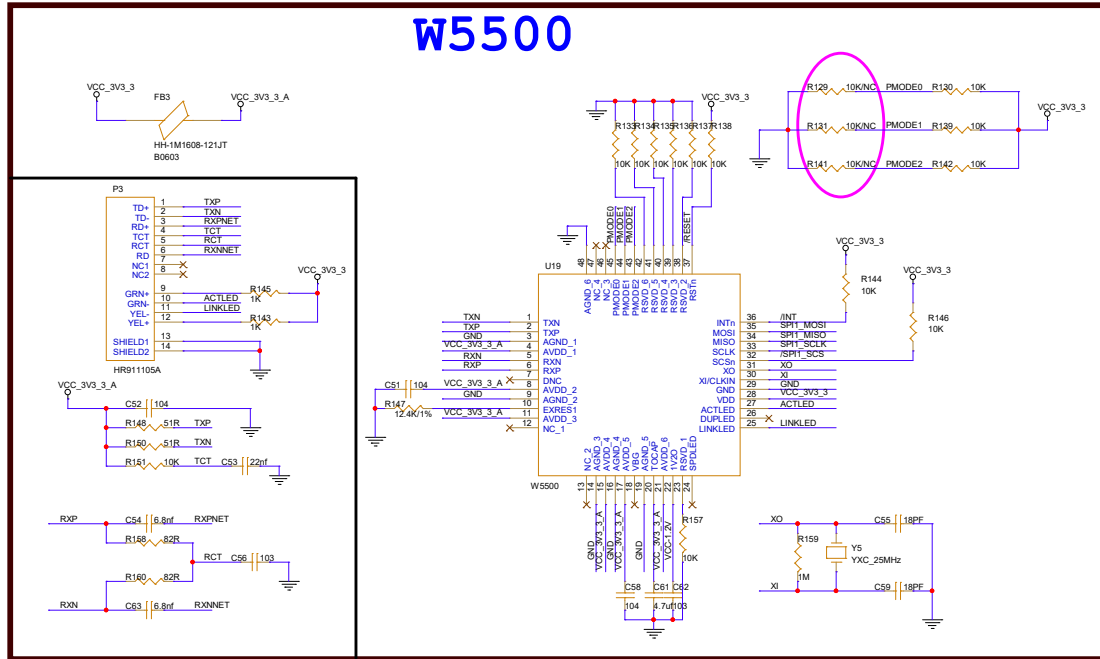
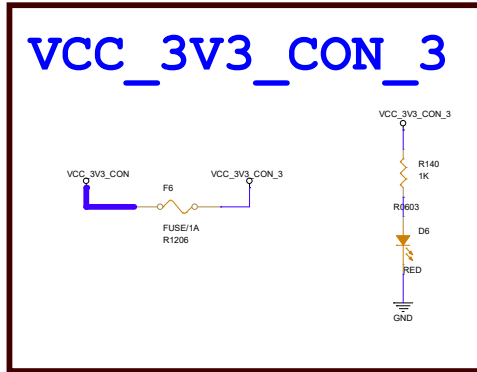


The diagram shows a circuit for VCC\_3V3\_3. It starts with a VCC\_3V3 input on the left, which connects to a 1A fuse (F5) and a 1K resistor (R132). The output of the fuse and resistor is labeled VCC\_3V3\_3. This output is connected to a 603 resistor (R603) and a red LED (D5). The LED is connected to ground (GND).

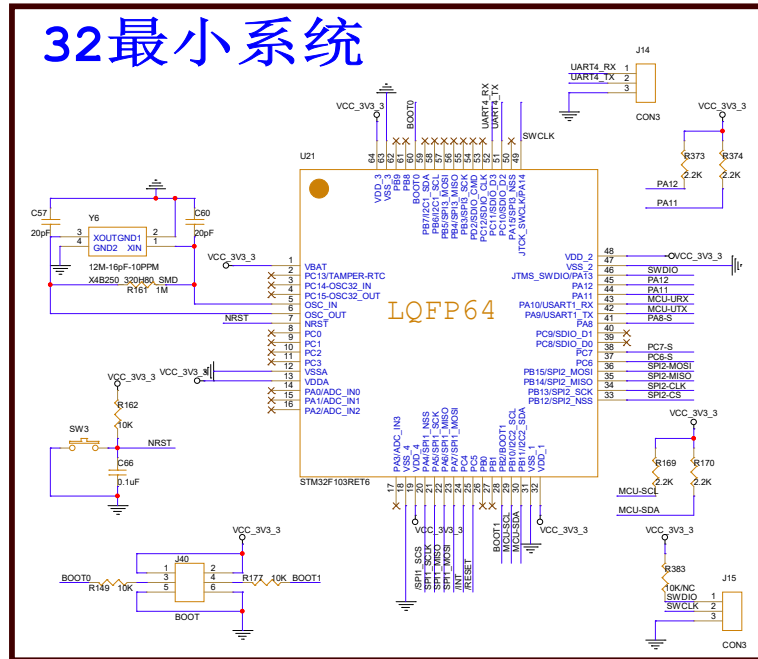
**VCC-12V**  
**转VCC\_3V3\_3**

# VCC\_3V3\_CON\_3

The diagram illustrates the electrical connection for a red LED (D6) powered by a 3V3 supply. The LED's anode is connected to the VCC\_3V3\_CON\_3 pin, which is also connected to a 1kΩ resistor (R140) and a fuse (F6). The LED's cathode is connected to ground through a 600Ω resistor (R603). The LED is labeled D6 and RED.

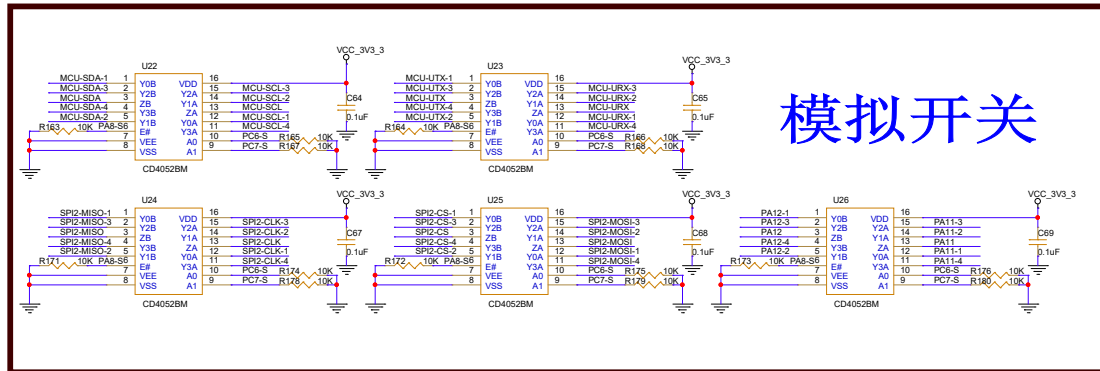


# 32最小系统

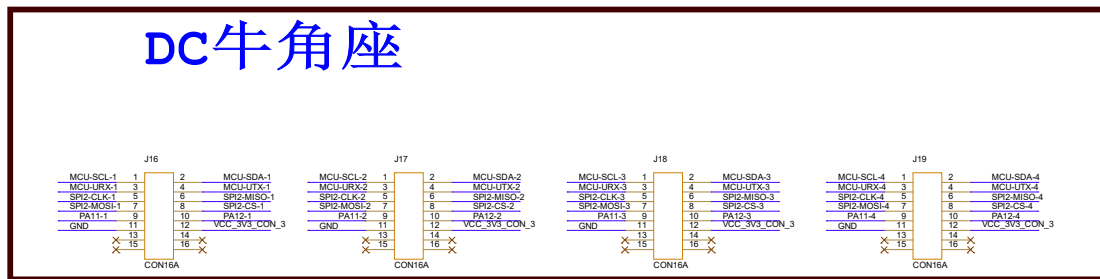


The image displays four circuit diagrams for connecting an MCU to an ADXL345 accelerometer using different communication protocols: I2C, SPI, and UART. Each diagram includes pin connections, decoupling capacitors, and pull-up resistors.

- Diagram 1 (I2C):** Shows an MCU (MCU-SDA-1) connected to an ADXL345 (CD4052BM) via I2C. The MCU's SDA (pin 1) and SCL (pin 2) are connected to the ADXL345's SDA (pin 16) and SCL (pin 15) respectively. A 10k pull-up resistor (R163) is connected to VCC\_3V3\_3 and the SDA line. A 1uF decoupling capacitor (C64) is connected to VCC\_3V3\_3 and the ADXL345's VDD (pin 16).
- Diagram 2 (SPI):** Shows an MCU (MCU-MISO-1) connected to an ADXL345 (CD4052BM) via SPI. The MCU's MISO (pin 1), MOSI (pin 2), and CS (pin 3) are connected to the ADXL345's MISO (pin 16), MOSI (pin 15), and CS (pin 14) respectively. A 10k pull-up resistor (R171) is connected to VCC\_3V3\_3 and the MISO line. A 1uF decoupling capacitor (C67) is connected to VCC\_3V3\_3 and the ADXL345's VDD (pin 16).
- Diagram 3 (UART):** Shows an MCU (MCU-UTX-1) connected to an ADXL345 (CD4052BM) via UART. The MCU's UTX (pin 1) and RX (pin 2) are connected to the ADXL345's UTX (pin 16) and RX (pin 15) respectively. A 10k pull-up resistor (R164) is connected to VCC\_3V3\_3 and the UTX line. A 1uF decoupling capacitor (C65) is connected to VCC\_3V3\_3 and the ADXL345's VDD (pin 16).
- Diagram 4 (I2C):** Shows an MCU (MCU-MISO-1) connected to an ADXL345 (CD4052BM) via I2C. The MCU's MISO (pin 1) and SDA (pin 2) are connected to the ADXL345's MISO (pin 16) and SDA (pin 15) respectively. A 10k pull-up resistor (R171) is connected to VCC\_3V3\_3 and the SDA line. A 1uF decoupling capacitor (C67) is connected to VCC\_3V3\_3 and the ADXL345's VDD (pin 16).



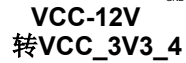
# DC牛角座





# VCC\_3V3\_4

VCC\_12V  
转VCC\_3V3\_4



# VCC\_3V3\_CON\_4

The diagram illustrates the electrical connection for the VCC\_3V3\_CON\_4 pin. It shows a series connection of three components: a 1K resistor (R200), a 6003 resistor (R6003), and a red LED (D8). The circuit is powered by VCC\_3V3\_CON and grounded (GND). A fuse (F8) is also shown in the path, with a rating of FUSE/1A and R1206.

```

graph TD
    VCC_3V3_CON[VCC_3V3_CON] --- F8[F8  
FUSE/1A  
R1206]
    F8 --- R200[R200  
1K]
    R200 --- R6003[R6003]
    R6003 --- D8[D8  
RED]
    D8 --- GND[GND]
  
```



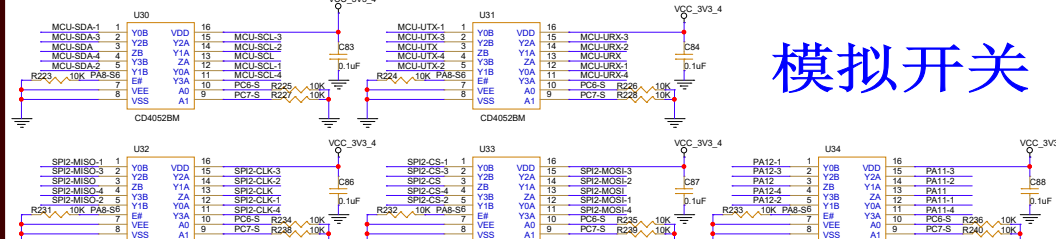
# 32最小系统



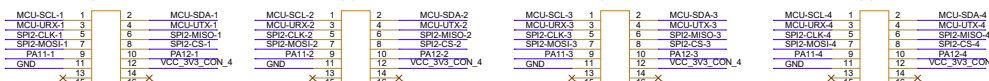
The image shows a detailed PCB layout for the W5500 module. The layout is organized into several sections:

- Top Section:** Features a BNC connector (FB4) connected to VCC\_3V3\_4 and VCC\_3V3\_4\_A. Below it is a component labeled HH-1M1608-121JT B0503.
- Bottom Section:** Features a 10-pin header (P4) with pins TD+, TD-, RD+, RD-, TCT, RCT, RXN, RXNET, NC2, and NC3. It is connected to VCC\_3V3\_4 and GND.
- Central Section:** Contains the W5500 chip. Its pins are connected to various components:
  - Pin 1 (TXN):** Connected to TXN.
  - Pin 2 (TXP):** Connected to TXP.
  - Pin 3 (GND):** Connected to GND.
  - Pin 4 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 5 (AVDD\_1):** Connected to AVDD\_1.
  - Pin 6 (RXN):** Connected to RXN.
  - Pin 7 (RXP):** Connected to RXP.
  - Pin 8 (DNC):** Connected to GND.
  - Pin 9 (AVDD\_2):** Connected to AVDD\_2.
  - Pin 10 (GND):** Connected to GND.
  - Pin 11 (EXRES1):** Connected to VCC\_3V3\_4\_A.
  - Pin 12 (AVDD\_3):** Connected to VCC\_3V3\_4\_A.
  - Pin 13 (NC3):** Connected to GND.
  - Pin 14 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 15 (GND):** Connected to GND.
  - Pin 16 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 17 (GND):** Connected to GND.
  - Pin 18 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 19 (GND):** Connected to GND.
  - Pin 20 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 21 (GND):** Connected to GND.
  - Pin 22 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 23 (GND):** Connected to GND.
  - Pin 24 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 25 (GND):** Connected to GND.
  - Pin 26 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 27 (GND):** Connected to GND.
  - Pin 28 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 29 (GND):** Connected to GND.
  - Pin 30 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 31 (GND):** Connected to GND.
  - Pin 32 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 33 (GND):** Connected to GND.
  - Pin 34 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 35 (GND):** Connected to GND.
  - Pin 36 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 37 (GND):** Connected to GND.
  - Pin 38 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 39 (GND):** Connected to GND.
  - Pin 40 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 41 (GND):** Connected to GND.
  - Pin 42 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 43 (GND):** Connected to GND.
  - Pin 44 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 45 (GND):** Connected to GND.
  - Pin 46 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 47 (GND):** Connected to GND.
  - Pin 48 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 49 (GND):** Connected to GND.
  - Pin 50 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 51 (GND):** Connected to GND.
  - Pin 52 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 53 (GND):** Connected to GND.
  - Pin 54 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 55 (GND):** Connected to GND.
  - Pin 56 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 57 (GND):** Connected to GND.
  - Pin 58 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 59 (GND):** Connected to GND.
  - Pin 60 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 61 (GND):** Connected to GND.
  - Pin 62 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 63 (GND):** Connected to GND.
  - Pin 64 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 65 (GND):** Connected to GND.
  - Pin 66 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 67 (GND):** Connected to GND.
  - Pin 68 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 69 (GND):** Connected to GND.
  - Pin 70 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 71 (GND):** Connected to GND.
  - Pin 72 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 73 (GND):** Connected to GND.
  - Pin 74 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 75 (GND):** Connected to GND.
  - Pin 76 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 77 (GND):** Connected to GND.
  - Pin 78 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 79 (GND):** Connected to GND.
  - Pin 80 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 81 (GND):** Connected to GND.
  - Pin 82 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 83 (GND):** Connected to GND.
  - Pin 84 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 85 (GND):** Connected to GND.
  - Pin 86 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 87 (GND):** Connected to GND.
  - Pin 88 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 89 (GND):** Connected to GND.
  - Pin 90 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 91 (GND):** Connected to GND.
  - Pin 92 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 93 (GND):** Connected to GND.
  - Pin 94 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 95 (GND):** Connected to GND.
  - Pin 96 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 97 (GND):** Connected to GND.
  - Pin 98 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
  - Pin 99 (GND):** Connected to GND.
  - Pin 100 (VCC\_3V3\_4\_A):** Connected to VCC\_3V3\_4\_A.
- Right Section:** Contains a crystal (Y7, 25MHz) and various capacitors (C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100).

The layout is color-coded with blue and red lines for different signal types. A large pink oval highlights a specific area of the top right, likely indicating a design change or a specific component placement.



# DC牛角座



VCC\_3V3\_5

VCC\_3V3

F8

VCC\_3V3\_5

R252  
1K

R603

D9

RED

GND

VCC-12V  
转VCC\_3V3\_5

# VCC\_3V3\_CON\_5

The schematic diagram illustrates the VCC\_3V3\_CON\_5 circuit. It begins with a power source labeled VCC\_3V3\_CON. The signal path continues through a fuse component, F10, which is specified as FUSE/1A and R1206. Following the fuse, the path leads to a resistor R260 with a value of 1K. The circuit then passes through a diode D10, which is a red LED. Finally, the circuit is connected to ground (GND). The output of this circuit is labeled VCC\_3V3\_CON\_5.

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[illegible]

# DC牛角座

VCC\_3V3\_6

VCC\_3V3

F11

VCC\_3V3\_6

R312 1K

R603

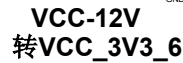
D11

RED

GND

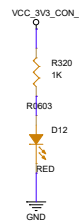
FUSE/1A  
R1206

VCC-12V  
转VCC\_3V3\_6



# VCC\_3V3\_CON\_6

The diagram illustrates the electrical connection between two power pins. On the left, a blue wire connects the **VCC\_3V3\_CON** pin to a blue pad. From this pad, a yellow wire passes through a component labeled **FUSE/1A R1206**. The wire then continues to a purple pad connected to the **VCC\_3V3\_CON\_6** pin. On the right, a vertical circuit path is shown, starting from **VCC\_3V3\_CON\_6** at the top, passing through a yellow resistor **R320 1K**, then a red LED **D12 RED** (pointing downwards), and finally a red resistor **R0603** connected to a **GND** symbol at the bottom.



The image shows a detailed PCB layout for the W5500 module. The layout is organized into several functional blocks:

- Top Section:** A voltage divider network (FB6) connected to VCC\_3V3\_6 and VCC\_3V3\_6\_A. A pull-up network (R300-R345) is connected to VCC\_3V3\_6 and the W5500's PMODE pins.
- Central Section:** The W5500 chip is shown with its pin connections to the microcontroller (U43) and other components. The W5500's pins are labeled with their functions (e.g., TXN, RXN, RXP, RXT, TXD, RXD, RSD, RSDV, RSDV2, RSDV3, RSDV4, RSDV5, RSDV6, RSDV7, RSDV8, RSDV9, RSDV10, RSDV11, RSDV12, RSDV13, RSDV14, RSDV15, RSDV16, RSDV17, RSDV18, RSDV19, RSDV20, RSDV21, RSDV22, RSDV23, RSDV24, RSDV25, RSDV26, RSDV27, RSDV28, RSDV29, RSDV30, RSDV31, RSDV32, RSDV33, RSDV34, RSDV35, RSDV36, RSDV37, RSDV38, RSDV39, RSDV40, RSDV41, RSDV42, RSDV43, RSDV44, RSDV45, RSDV46, RSDV47, RSDV48, RSDV49, RSDV50, RSDV51, RSDV52, RSDV53, RSDV54, RSDV55, RSDV56, RSDV57, RSDV58, RSDV59, RSDV60, RSDV61, RSDV62, RSDV63, RSDV64, RSDV65, RSDV66, RSDV67, RSDV68, RSDV69, RSDV70, RSDV71, RSDV72, RSDV73, RSDV74, RSDV75, RSDV76, RSDV77, RSDV78, RSDV79, RSDV80, RSDV81, RSDV82, RSDV83, RSDV84, RSDV85, RSDV86, RSDV87, RSDV88, RSDV89, RSDV90, RSDV91, RSDV92, RSDV93, RSDV94, RSDV95, RSDV96, RSDV97, RSDV98, RSDV99, RSDV100, RSDV101, RSDV102, RSDV103, RSDV104, RSDV105, RSDV106, RSDV107, RSDV108, RSDV109, RSDV110, RSDV111, RSDV112, RSDV113, RSDV114, RSDV115, RSDV116, RSDV117, RSDV118, RSDV119, RSDV120, RSDV121, RSDV122, RSDV123, RSDV124, RSDV125, RSDV126, RSDV127, RSDV128, RSDV129, RSDV130, RSDV131, RSDV132, RSDV133, RSDV134, RSDV135, RSDV136, RSDV137, RSDV138, RSDV139, RSDV140, RSDV141, RSDV142, RSDV143, RSDV144, RSDV145, RSDV146, RSDV147, RSDV148, RSDV149, RSDV150, RSDV151, RSDV152, RSDV153, RSDV154, RSDV155, RSDV156, RSDV157, RSDV158, RSDV159, RSDV160, RSDV161, RSDV162, RSDV163, RSDV164, RSDV165, RSDV166, RSDV167, RSDV168, RSDV169, RSDV170, RSDV171, RSDV172, RSDV173, RSDV174, RSDV175, RSDV176, RSDV177, RSDV178, RSDV179, RSDV180, RSDV181, RSDV182, RSDV183, RSDV184, RSDV185, RSDV186, RSDV187, RSDV188, RSDV189, RSDV190, RSDV191, RSDV192, RSDV193, RSDV194, RSDV195, RSDV196, RSDV197, RSDV198, RSDV199, RSDV200, RSDV201, RSDV202, RSDV203, RSDV204, RSDV205, RSDV206, RSDV207, RSDV208, RSDV209, RSDV210, RSDV211, RSDV212, RSDV213, RSDV214, RSDV215, RSDV216, RSDV217, RSDV218, RSDV219, RSDV220, RSDV221, RSDV222, RSDV223, RSDV224, RSDV225, RSDV226, RSDV227, RSDV228, RSDV229, RSDV230, RSDV231, RSDV232, RSDV233, RSDV234, RSDV235, RSDV236, RSDV237, RSDV238, RSDV239, RSDV240, RSDV241, RSDV242, RSDV243, RSDV244, RSDV245, RSDV246, RSDV247, RSDV248, RSDV249, RSDV250, RSDV251, RSDV252, RSDV253, RSDV254, RSDV255, RSDV256, RSDV257, RSDV258, RSDV259, RSDV260, RSDV261, RSDV262, RSDV263, RSDV264, RSDV265, RSDV266, RSDV267, RSDV268, RSDV269, RSDV270, RSDV271, RSDV272, RSDV273, RSDV274, RSDV275, RSDV276, RSDV277, RSDV278, RSDV279, RSDV280, RSDV281, RSDV282, RSDV283, RSDV284, RSDV285, RSDV286, RSDV287, RSDV288, RSDV289, RSDV290, RSDV291, RSDV292, RSDV293, RSDV294, RSDV295, RSDV296, RSDV297, RSDV298, RSDV299, RSDV300, RSDV301, RSDV302, RSDV303, RSDV304, RSDV305, RSDV306, RSDV307, RSDV308, RSDV309, RSDV310, RSDV311, RSDV312, RSDV313, RSDV314, RSDV315, RSDV316, RSDV317, RSDV318, RSDV319, RSDV320, RSDV321, RSDV322, RSDV323, RSDV324, RSDV325, RSDV326, RSDV327, RSDV328, RSDV329, RSDV330, RSDV331, RSDV332, RSDV333, RSDV334, RSDV335, RSDV336, RSDV337, RSDV338, RSDV339, RSDV340, RSDV341, RSDV342, RSDV343, RSDV344, RSDV345, RSDV346, RSDV347, RSDV348, RSDV349, RSDV350, RSDV351, RSDV352, RSDV353, RSDV354, RSDV355, RSDV356, RSDV357, RSDV358, RSDV359, RSDV360, RSDV361, RSDV362, RSDV363, RSDV364, RSDV365, RSDV366, RSDV367, RSDV368, RSDV369, RSDV370, RSDV371, RSDV372, RSDV373, RSDV374, RSDV375, RSDV376, RSDV377, RSDV378, RSDV379, RSDV380, RSDV381, RSDV382, RSDV383, RSDV384, RSDV385, RSDV386, RSDV387, RSDV388, RSDV389, RSDV390, RSDV391, RSDV392, RSDV393, RSDV394, RSDV395, RSDV396, RSDV397, RSDV398, RSDV399, RSDV400, RSDV401, RSDV402, RSDV403, RSDV404, RSDV405, RSDV406, RSDV407, RSDV408, RSDV409, RSDV410, RSDV411, RSDV412, RSDV413, RSDV414, RSDV415, RSDV416, RSDV417, RSDV418, RSDV419, RSDV420, RSDV421, RSDV422, RSDV423, RSDV424, RSDV425, RSDV426, RSDV427, RSDV428, RSDV429, RSDV430, RSDV431, RSDV432, RSDV433, RSDV434, RSDV435, RSDV436, RSDV437, RSDV438, RSDV439, RSDV440, RSDV441, RSDV442, RSDV443, RSDV444, RSDV445, RSDV446, RSDV447, RSDV448, RSDV449, RSDV450, RSDV451, RSDV452, RSDV453, RSDV454, RSDV455, RSDV456, RSDV457, RSDV458, RSDV459, RSDV460, RSDV461, RSDV462, RSDV463, RSDV464, RSDV465, RSDV466, RSDV467, RSDV468, RSDV469, RSDV470, RSDV471, RSDV472, RSDV473, RSDV474, RSDV475, RSDV476, RSDV477, RSDV478, RSDV479, RSDV480, RSDV481, RSDV482, RSDV483, RSDV484, RSDV485, RSDV486, RSDV487, RSDV488, RSDV489, RSDV490, RSDV491, RSDV492, RSDV493, RSDV494, RSDV495, RSDV496, RSDV497, RSDV498, RSDV499, RSDV500, RSDV501, RSDV502, RSDV503, RSDV504, RSDV505, RSDV506, RSDV507, RSDV508, RSDV509, RSDV510, RSDV511, RSDV512, RSDV513, RSDV514, RSDV515, RSDV516, RSDV517, RSDV518, RSDV519, RSDV520, RSDV521, RSDV522, RSDV523, RSDV524, RSDV525, RSDV526, RSDV527, RSDV528, RSDV529, RSDV530, RSDV531, RSDV532, RSDV533, RSDV534, RSDV535, RSDV536, RSDV537, RSDV538, RSDV539, RSDV540, RSDV541, RSDV542, RSDV543, RSDV544, RSDV545, RSDV546, RSDV547, RSDV548, RSDV549, RSDV550, RSDV551, RSDV552, RSDV553, RSDV55



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